

From serial MRI to objective tumor tracking.

3D segmentation turns follow-up scans into lesion volumes that can be compared over time.

MRI DATA STANDARD Presentation reference · BraTS-standardized input

T1 · T1c · T2 · FLAIR
MRI sequences

240 × 240 × 155
reference matrix

1.0 mm³
voxel resolution

NIfTI + geometry
spacing & orientation preserved

HOW IT WORKS One clear path from scan to insight

IMPLEMENTED scan analysis



SERIAL MRI EVIDENCE

**One tumor.
Five timepoints.**

Segmentation overlays show the same lesion across follow-up scans.



EXAMPLE CASE · MIDTERM PROTOTYPE

Tumor volume across five MRI timepoints

14.8 -> 1.3 cm³

Illustrative case result

01 3D segmentation
nnU-Net produces a 3D mask per scan.

02 Objective volume tracking
Change measured in cm³ over time.

03 Quality first
Standardize + skull-strip to reduce false positives; QC + human review.